Per-operator cross-environment prediction — results

Generated by `scripts/analysis/operator_model_evaluation.py` (`make operator-results`). Raw numbers: `results/operator_model_metrics.json`, per-pair predictions: `results/operator_model_pairs.csv`.

Setup

environments	**8** container cells, CPU quota {1,2,4,8} × memory {1g,3g}, crossed
workload	22 TPC-H queries at SF1, 4 repetitions, first discarded as warmup
pairs	**1232** ordered dev→prod (8×7 machine pairs × 22 queries)
plans	**0 of 22** queries chose a different plan across cells
capability	`operator_rates_v2`: 5 serial axes + `parallel_speedup`, probed in SQL per cell

Why containers. CPU quota × memory quota are orthogonal *by construction*. Renting more EC2 instance types cannot buy that: instance types are correlated bundles, which is why the five-machine EC2 signature correlates at r = +0.958 with 97.9% of variance on one component.

What the identical plans mean. Resource quotas do not change the planner's cost parameters, so the plan is stable and this grid **isolates the capability term cleanly**. The corollary is a limitation: with no plan divergence here, **move 3 (take the plan from production) cannot be evaluated on this grid at all.**

Results

predictor	MAPE %	MdAPE %	q-error p50	q-error p95
naive (assume prod = dev)	72.52	36.78	1.457	4.586
`scalar_seq` (one probe axis)	73.63	37.08	1.458	4.560
`scalar_geo` (geomean of serial axes)	71.46	35.95	1.451	4.519
`operator_serial` (per-operator, no parallelism axis)	71.11	35.99	1.459	4.520
`scalar_geo_par` (one scalar + parallelism, applied whole-query)	29.14	20.22	1.232	2.396
`scalar_localized_par` (one scalar + parallelism, **localised to the parallel part**)	**22.23**	18.19	1.197	1.700
`operator` (per-operator + localised parallelism)	**22.30**	18.13	1.196	1.688

Paired cluster bootstrap on the MAPE difference, resampling whole production machines (positive ⇒ the second predictor is better):

comparison	Δ MAPE	95% CI	
naive – operator	+50.21 pp	[+39.81, +61.75]	**significant**
scalar_seq – operator	+51.33 pp	[+40.61, +63.22]	**significant**
scalar_geo – operator	+49.15 pp	[+38.95, +60.35]	**significant**
operator_serial – operator	+48.80 pp	[+38.60, +60.13]	**significant**
scalar_geo_par – operator	+6.83 pp	[+0.81, +14.48]	**significant**
scalar_geo_par – scalar_localized_par	**+6.91 pp**	**[+0.95, +14.73]**	**significant**
scalar_localized_par – operator	**−0.08 pp**	**[−0.73, +0.44]**	**not significant**

What this establishes

1. The predictor works: 22.30% against 72.52% for doing nothing, +50.21 pp, significant on 1232 pairs.

2. A capability signature with no parallelism term is worse than, or no better than, doing nothing. `scalar_seq` 73.63%, `scalar_geo` 71.46% and `operator_serial` 71.11% all sit level with or above naive's 72.52%. This reproduces, under controlled conditions, exactly the failure the original signature showed laptop→EC2 (324% against 123% for naive), and the mechanism is now identified rather than guessed: **a CPU quota does not slow a single-threaded operator down**, so the serial axes barely move across the grid while the parallel speedup moves 3.66×.

axis	spread across the 8 cells
sort	1.02×
seqscan	1.03×
nestloop	1.10×
hashagg	1.12×
hashjoin	1.13×
parallel_speedup	**3.66×**

3. The gain comes from plan structure, NOT from per-operator capability. This is the ablation that matters. `scalar_localized_par` uses a *single* serial capability scalar for the whole plan and differs from `operator` only in that it does not weight per operator. It scores **22.23% against the full model's 22.30% — a difference of −0.08 pp with a CI of [−0.73, +0.44].** They are statistically indistinguishable.

Meanwhile, applying the parallelism correction to the *whole* query rather than localising it to the nodes below a `Gather` costs **+6.91 pp (CI [+0.95, +14.73], significant)**.

So the contribution is: **knowing which part of the plan runs in parallel, and correcting only that part.** Per-operator rate differentiation adds nothing measurable here — and the dispersion table says why, since the serial axes span only 1.02–1.13×. Testing per-operator rates on their merits needs

environments whose operator rates genuinely differ, which means different hardware (a second ISA, cache hierarchy or memory subsystem), not a quota.

This is a simplification result, and it should be reported as one. A shippable model needs one serial capability scalar, one parallel speedup, and the plan's parallel/serial wall-time split. The five-axis probe is how the speedup is measured and may matter across ISAs, but on this evidence its per-axis detail is not carrying the prediction.

4. The new signature is genuinely multi-dimensional.

	old 2-scalar signature	operator rates
PC1 share of variance	97.9%	**59.4%**
mean |	r|	between axes
effective rank at 95% variance	1	**4**

This was the blocker that forced every negative result in Tier 1, and it is resolved — though note the previous point: multi-dimensional does not yet mean *usefully* multi-dimensional.

Limits

- **Few-shot calibration buys almost nothing here** (28.54 → 21.1–21.3% at k=8, and k=1 by `spread` is *worse*). Expected: the cells are homogeneous, so there is no per-machine bias to remove. Move 4 is evaluated properly in laptop_to_prod.md, where there is one.
- **Two of eight cells failed the probe repeatability gate** (`c2m1g` , `c4m1g` ; cv 0.165 and 0.195). They are kept rather than dropped, because discarding them would bias the grid toward the cells that happened to be measured in quiet moments. The reliable subset (6 cells, 660 pairs) shows the same pattern more strongly: operator **23.19%**, `scalar_localized_par` 23.81%, `scalar_geo_par` 32.77%, naive 85.68%.
- **Same host, same ISA**; single client; identical SF1 data, so the cardinality term is exercised only at ratio 1.0; 90.9% of pairs involve a parallel plan, so the parallelism axis is doing most of the work by construction of this workload.

Honest one-line summary

Parallelism is the result, and localising it to the parallel part of the plan is the refinement. Together they turn a predictor level with doing nothing (71–74%) into one that beats it decisively (22.30% against 72.52%). Per-operator rate weighting, the extra axes and few-shot calibration are built and mechanically verified, but on this grid only the parallelism axis and its plan-structure localisation are shown to matter.